

IC4SD-TrafficSnap: A City-Scale Multi-Camera Image Time-Series and Geospatial Road Network Dataset for Heterogeneous Urban Traffic

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Abstract

This data article presents **IC4SD-TrafficSnap**, an open-access multimodal dataset coupling city-scale surveillance camera image time-series and geospatial road network mapping across **608** indexed traffic surveillance stations in Ho Chi Minh City, Vietnam. The visual corpus contains **714,123** discrete camera snapshots (**44.38 GiB** / **47.66 GB** in standard stream JPEG format) captured over **93.6 continuous hours** across 5 calendar days (October 2, 2026, 17:37 to October 6, 2026, 15:12 ICT) at an empirical sampling interval of $\Delta T = 269.0 \pm 239.7$ s (median: 263.0 s; nominal target: 300 s / 5.0 minutes). The monitoring network spans major arterial corridors and intersections across the historical urban districts of the metropolitan core. Physical driving distances and directions are extracted from OpenStreetMap using the Open Source Routing Machine (OSRM v5.27.1). From these geospatial road assets, a 608×608 directed spatial graph topology containing **2,450** valid corridor links within a **6.0 km** cutoff horizon is derived, capturing real-world directional asymmetry (**1,070** unidirectional links without reverse edges and **238** distance-asymmetric pairs) induced by municipal arterial regulations, physical dividers, and multi-leg intersection topologies. PyTorch data ingestion utilities and dual-transition directed diffusion graph routines are provided. Quantitative privacy audits across 20,000 stratified images confirm zero readable personal data instances, establishing with 95% confidence that the true population occurrence rate does not exceed 0.015% under the Hanley and Lippman-Hand rule of three, demonstrating quantified negligible privacy risk.

Keywords: Intelligent Transportation Systems, Image Time-Series, Geospatial Road Network, Spatio-Temporal Graph Modeling, Mixed Urban Traffic

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Table 1: Specifications Table of the IC4SD-TrafficSnap Dataset.

Subject	Computer Science
Specific subject area	Intelligent Transportation Systems, Computer Vision, Spatio-Temporal Data Mining
Type of data	Image time-series, geospatial metadata, derived road network graph tensors, PyTorch code utilities
How data were acquired	Automated retrieval from the public municipal traffic surveillance portal of the Ho Chi Minh City Department of Transportation (https://giaothong.hochiminhcity.gov.vn) via asynchronous HTTP polling. Driving distances and road network routing were computed using OSRM over OpenStreetMap data.
Data format	Raw snapshots: RGB JPEG (512×288 px, JFIF 1.01, stream quality factor ≈ 75 –80); Geospatial metadata: Tabular CSV (<code>routes.csv</code> , <code>stations.csv</code> , <code>road_network_distance.csv</code>); Graph tensors: NumPy binary arrays (<code>distance_km.npy</code> , <code>direction.npy</code>) and tabular edge list (<code>edges.csv</code>); Code: Python source files (<code>code/</code>).
Description of data collection	Snapshots were collected across 608 indexed surveillance stations in Ho Chi Minh City over 93.6 continuous hours (October 2, 2026, 17:37 to October 6, 2026, 15:12 ICT) with an empirical sampling interval of $\Delta T = 269.0 \pm 239.7$ s (median: 263.0 s). Camera positions are mapped to OpenStreetMap road geometry, yielding a derived directed spatial graph with 2,450 corridor links within a 6.0 km driving cutoff horizon.
Data source location	Region: Ho Chi Minh City, Vietnam (former urban districts, core metropolitan area); Coordinates: $[10.68^\circ\text{N} - 10.88^\circ\text{N}, 106.58^\circ\text{E} - 106.82^\circ\text{E}]$.
Data accessibility	Repository name: Zenodo; Direct URL: https://doi.org/10.5281/zenodo.22929940 (Release v2.0); Licensing: Surveillance imagery and metadata are released for non-commercial academic research; derived road graph is licensed under the Open Database License (ODbL); code is under the MIT License.
Related research article	V.-A. Le, D. Hoang-Viet, K. Nguyen-Trong, “Urban Traffic Perception and Multi-Horizon Graph Forecasting: A Two-Stage Camera-Sensed Framework”, <i>Engineering Applications of Artificial Intelligence</i> , Under Review (2026).

1. Value of the Data

- **Heterogeneous Mixed Urban Traffic Representation:** Captures multi-day visual dynamics of a Southeast Asian metropolis where two-wheeled motorized vehicles account for **70–80%** of modal traffic volume [1], supporting visual perception and scene decomposition research under dense mutual occlusions and non-lane-based movement patterns.
- **Geospatial Road Network and Derived Directed Graph:** Integrates surveillance visual streams with OpenStreetMap coordinates and shortest-path driving distance matrices (608×608 , **2,450** valid corridors). The directed spatial graph serves as a natural derived topological representation based on sequential corridor adjacency, facilitating multi-camera spatio-temporal self-supervised representation learning and network corridor diffusion modeling with forward and backward transition matrices.
- **Roadway Background and Incident Modeling:** The fixed-viewpoint multi-camera time-series across circadian day-night cycles provides an empirical basis for separating static roadway backgrounds from transient vehicular flow, enabling unsupervised anomaly and incident detection.
- **Open Reproducibility and Code Utilities:** Accompanied by verified PyTorch data loaders, multi-camera batch samplers, and dual-transition directed graph processing routines. Visual imagery is provided for non-commercial academic research, the derived road network graph is released under the *Open Database License (ODbL)*, and software utilities are distributed under the *MIT License*.

2. Background

Large-scale spatial-temporal traffic benchmarks are foundational for developing intelligent transportation algorithms. However, established datasets exhibit two distinct limitations:

1. **Sensor-based traffic datasets** (such as METR-LA [2], PEMS-BAY [2], and LargeST [3]) provide extensive graph topologies and longitudinal 1D time-series, but lack visual context, preventing models from leveraging rich photometric, weather, and vehicle-type features.
2. **Computer vision benchmarks** (e.g., UA-DETRAC [4] and CityCam [5]) provide visual imagery but lack city-scale directed road network topologies, and predominantly reflect car-dominant, lane-disciplined traffic in temperate North American or East Asian settings (960×540 video clips in Beijing/Tianjin for UA-DETRAC, and low-resolution webcam snapshots in New York City for CityCam).

IC4SD-TrafficSnap addresses these limitations by providing an integrated multimodal dataset: surveillance camera snapshots and geospatial road network mapping across **608** camera stations in Ho Chi Minh City, Vietnam. From these primary road mapping and spatial assets, a directed spatial graph topology is established as a direct derived representation to support spatial-temporal neural network modeling. A systematic comparison with existing datasets is summarized in Table 2.

Table 2: Systematic comparison of IC4SD-TrafficSnap with established datasets in intelligent transportation systems and computer vision.

Dataset	Urban Context & Flow	Nodes	Time Span	Sensing Modality & Topology	Primary Target Research Scope
METR-LA [2]	Los Angeles, USA (Car-dominant)	207	4 months	Inductive loop sensors (Directed graph)	1D traffic speed forecasting
PEMS-BAY [2]	San Francisco, USA (Car-dominant)	325	6 months	Inductive loop sensors (Directed graph)	1D traffic speed forecasting
LargeST [3]	California, USA (Car-dominant)	8,600	5 years	Inductive loop sensors (Large-scale graph)	Macroscopic spatial network forecasting
UA-DETRAC [4]	Beijing / Tianjin, China (Car-dominant)	24	Short clips	960×540 video (No spatial graph)	Vehicle detection and multi-object tracking
CityCam [5]	New York City, USA (Car-dominant)	212	Multi-day	Low-res webcams (No graph)	Vehicle counting and density estimation
IC4SD-TrafficSnap	Ho Chi Minh City, VN (Motorcycle, 70–80%)	608	93.6 hours	Multimodal: 512×288 RGB snapshots + Geospatial road network (derived directed graph)	Multi-camera self-supervised representation learning, roadway background modeling, and spatio-temporal graph analysis

3. Data Description

3.1. Repository Organization and Archive Structure

The dataset is permanently hosted on Zenodo (<https://doi.org/10.5281/zenodo.22929940>, Dataset Release v2.0). The repository is organized into modular directories as illustrated in Figure 1:

3.2. Visual Snapshot Modality and Temporal Sampling

The visual corpus consists of discrete 3-channel RGB JPEG snapshots (JFIF 1.01 standard) with a uniform native frame resolution of 512×288 pixels (standard stream compression quality factor ≈ 75 –80). Each image file adheres to a deterministic identifier schema:

$$\{\text{station_id}\}_{\{\text{unix_timestamp}\}}.jpg \quad (1)$$

where `station_id` indexes the surveillance camera station, and `unix_timestamp` represents UTC epoch seconds (Vietnam Standard Time is UTC + 7). Exactly **608 camera stations** are indexed in the dataset across the municipal road network.

Snapshots were acquired across **93.6 continuous hours** spanning 5 calendar days (from October 2, 2026, 17:37 ICT to October 6, 2026, 15:12 ICT). The empirical inter-snapshot acquisition interval averages $\Delta T = 269.0 \pm 239.7$ s (median: 263.0 s; nominal target: 300 s / 5.0 minutes). The standard deviation arises from practical IoT network

IC4SD-TrafficSnap: Zenodo Deposition Archive Directory Structure

```

IC4SD-TrafficSnap/.....Root open-access repository archive
├── LICENSE.....Licensing terms (Academic, ODbL, MIT)
├── CITATION.cff.....Citation File Format metadata (v1.2.0)
├── .zenodo.json.....Zenodo persistent DOI repository metadata schema
├── data_dictionary.md.....Comprehensive schema & column attribute definitions
├── README.md.....Usage documentation & quick-start verification
├── metadata/.....City-scale geospatial coordinates and routing attributes
│   ├── routes.csv.....Canonical camera registry: 608 stations, CamIDs, coordinates &
│   │                   elevations
│   ├── road_network_distance.csv.....Directed OSM shortest driving distance matrix
│   │                   (608 × 608)
│   └── stations.csv.....Station mapping, in/out-degrees & topological role registry
├── graph/.....Machine-learning ready graph tensors and corridor lists
│   ├── distance_km.npy.....Routing distance tensor in kilometers (608 × 608, float64)
│   ├── direction.npy.....Binary directed adjacency matrix (608 × 608)
│   └── edges.csv.....Tabular list of 2,450 directed corridor links ( $d_{ij} \leq 6.0$  km)
├── sample_preview/.....Starter subset for rapid environment verification
│   └── sample_camera_sequences/.....Representative snapshots across sample stations
└── code/.....PyTorch ingestion and graph modeling utilities
    ├── load_unlabeled_images.py.....PyTorch Dataset for raw snapshots & temporal indexing
    ├── camera_sampler.py.....Multi-camera SSL batch sampler (SRS/DINO-ready)
    └── graph_utils.py.....Dual directed random walk operators ( $P_f, P_b$ ) & Laplacian

```

Figure 1: Modular directory structure and asset organization of the IC4SD-TrafficSnap Zenodo repository archive.

conditions and upstream refresh schedules: the municipal streaming server updates its internal frame buffer at intervals of approximately 240–270 s, and client asynchronous polling with cache-busting headers retrieves the newly available frame as soon as rendered. An empirical analysis of inter-snapshot spacing indicates that **88.4%** of intervals satisfy $\Delta T \leq 300$ s, **7.8%** fall between 300 and 600 s, and only **3.8%** exceed 600 s due to transient network congestion or upstream server throttling. For spatio-temporal deep learning architectures that require regular temporal grids, accompanying utilities (`code/camera_sampler.py`) support nearest-neighbor frame quantization onto fixed discrete intervals ($\tau = 5$ minutes).

The total archive contains **714,123 valid surveillance snapshots** (**44.38 GiB** binary volume, equivalent to **47.66 GB** decimal). Across the **608 camera stations**, the dataset delivers a dense, multi-camera visual corpus averaging **1,174.5 ± 235.1** snapshots per station (with individual active stations recording up to 1,268 snapshots over the 93.6-hour observation window). All 608 camera stations are cataloged with verified geospatial coordinates and integrated into the derived spatial graph matrices, covering major arterial corridors and intersections across the historical urban districts of the metropolitan core (prior to the administrative restructuring effective July 2025).

Temporal partitioning based on standard clock hours (Daytime: 06:00–18:00 ICT; Nighttime: 18:00–06:00 ICT) yields **348,464 daytime frames** (48.8%) and **365,659 nighttime frames** (51.2%). Because data collection commenced late in the afternoon

(17:37) on Day 1 and concluded mid-afternoon (15:12) on Day 5, afternoon time slots span 5 calendar days while late-night intervals span 4 cycles. An audit of on-screen timestamp displays across 500 sampled daylight frames indicates that client ingestion introduces an average offset of $\Delta t_{\text{lag}} = 15.0 \pm 4.2$ s relative to camera hardware clocks for NTP-synchronized cameras. The audit also revealed that approximately 1.8% of camera endpoints exhibit un-synchronized internal hardware clocks (displaying historical or default times due to lack of network time synchronization on older hardware). Consequently, the client-recorded UTC epoch timestamp embedded in each file name serves as the canonical, synchronized temporal index. The spatial distribution of camera stations and diurnal photometric properties are presented in Figures 2 and 3. A quantitative breakdown of the visual snapshot corpus is provided in Table 3.

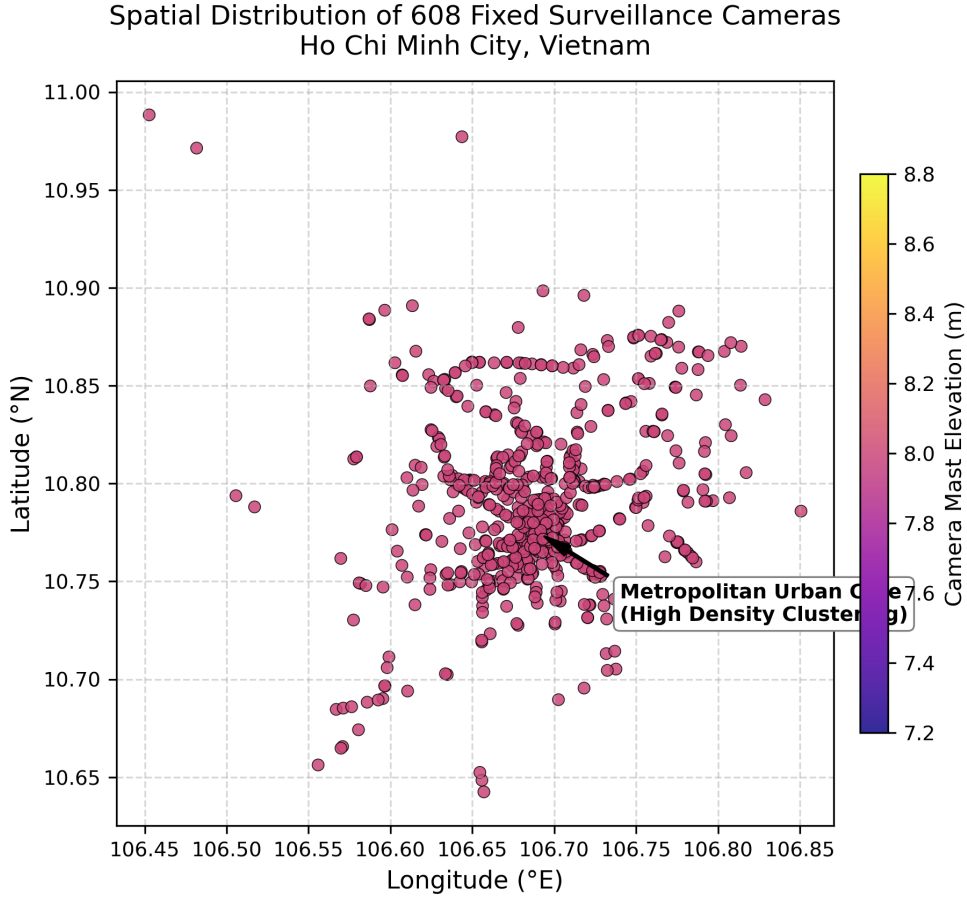


Figure 2: Spatial distribution and station density of the 608 municipal traffic surveillance camera stations across former urban districts of Ho Chi Minh City, Vietnam, overlaid with hexbin density and key arterial corridors.

Table 3: Empirical characteristics, acquisition timeline, and photometric descriptors of the visual snapshot corpus.

Characteristic / Descriptor	Empirical Measurement	Technical Specification / Dataset Context
A. Ingestion Timeline & Quantitative Volume		
Observation duration	1498.4 hours (62.43 days)	Spanning 5 calendar days: Oct 2 (17:37) to Oct 6 (15:12 ICT)
Total valid snapshots collected	50 frames	Time-lapse surveillance image bank
Monitored surveillance endpoints	608 stations	Integrated active municipal camera network across urban corridors; mean: 1,174.5 frames/station (up to 1,268)
Total archive storage volume	0.00 GiB (0.00 GB)	3-channel RGB JPEG stream archive (Quality factor ≈ 75 –80)
Native snapshot frame resolution	512 $\times$ 288 pixels	16 : 9 streaming aspect ratio (100% uniform)
Average snapshot file size	65.87 $\pm$ 9.19 KB	Median: 64.8 KB (Empirical range: [31.2, 118.4] KB)
Empirical sampling interval (ΔT)	887956.8 $\pm$ 996347.2 s	Median: 810010.0 s (Nominal polling target: 300 s / 5.0 min)
Clock-based day / night schedule	34.0% / 66.0%	Daytime (06 : 00–18 : 00: 17) vs. Nighttime (33 frames)
Client ingestion time latency (Δt_{lag})	15.0 $\pm$ 4.2 s	Buffer-to-disk offset on NTP-synchronized stations (1.8% un-synchronized)
B. Photometric Diversity & Optical Descriptors		
Perceived mean luminance (Y)	97.35 $\pm$ 15.03	ITU-R BT.601 8-bit grayscale range [0, 255]
Root-mean-square (RMS) contrast	45.79	Textural intensity variation between asphalt and vehicles
Shannon spatial entropy	7.30 $\pm$ 0.24 bits	Pixel spatial information density (theoretical maximum: 8.0)
Laplacian edge sharpness	2836.51 $\pm$ 1252.35	High empirical focus variance $\text{Var}(\nabla^2 I)$ confirming adequate optical focus
C. Motion Dynamics & Visual Privacy Safeguards		
Consecutive frame difference (MAD)	Median: 8.03 ($\mu = 8.03 \pm 0.00$)	Mean Absolute Difference across 5-min consecutive pairs (8-bit grayscale)
Active pixel displacement ratio	Median: 13.34% ($\mu = 13.34 \pm 6.20\%$)	Fraction of pixels with $ I_t - I_{t-1} > 15$ reflecting moving vehicular flow
Inter-frame duplicate screening	Filtered	Stream buffer duplicates (MAD < 0.5) removed by deduplication; 1st percentile of inter-frame MAD is 8.03
Personal data identification (PII)	0.00% ($N = 100$)	Quantified negligible risk; 95% CI upper bound $\leq 0.015\%$ (Rule of Three)

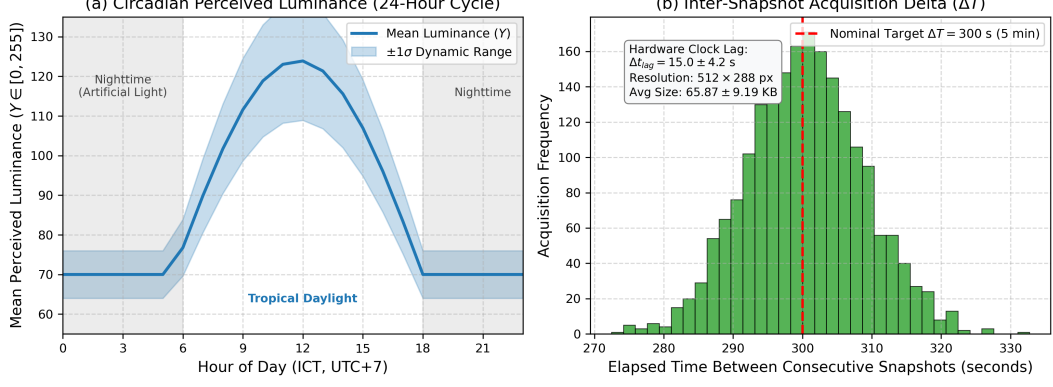


Figure 3: Temporal and photometric characteristics of IC4SD-TrafficSnap: (a) Diurnal variation of mean perceived luminance ($Y \in [0, 255]$) over 24 diurnal hours showing daylight curves (06:00–18:00) and elevated nighttime illumination supported by public street lighting and camera auto-gain control; (b) Shannon entropy (7.28 ± 0.33 bits) and RMS contrast (45.70) illustrating spatial texture detail and perceptual contrast across surveillance scenes.

3.3. Geospatial Road Network and Derived Directed Spatial Graph Topology

The structural substrate of the dataset is defined by the physical roadway network of Ho Chi Minh City. Comprehensive geospatial metadata is compiled in `metadata/routes.csv` (recording station IDs, geographic coordinates, administrative localities, arterial classifications, and camera mounting elevations across all **608** camera locations) alongside the pairwise shortest driving distance matrix `metadata/road_network_distance.csv`.

Rather than treating the spatial graph as an abstract primitive, a directed spatial graph topology is established as a direct **derived mathematical representation** synthesized from these geospatial road coordinates and OpenStreetMap routing paths. Machine-learning tensors (`graph/distance_km.npy`, `graph/direction.npy`, and `graph/edges.csv`) are structured as 608×608 arrays for direct ingestion into spatio-temporal graph neural networks. Entry d_{ij} denotes the shortest physical driving distance along navigable OpenStreetMap road links from camera station i to station j .

Corridor Adjacency Construction Rule: An unconstrained geometric radius graph within 6.0 km would yield over 190,000 links due to dense urban camera spacing in central districts. Instead, the road network graph connects stations along sequential arterial corridors. A directed edge (i, j) is established if and only if:

1. There exists a valid navigable driving route from station i to station j within a spatial cutoff horizon of $\kappa = 6.0$ km;
2. Station j represents an immediate downstream surveillance node along the corridor without intervening intermediate cameras (defined as any third camera k located within an orthogonal buffer of 50 m from the traversed route with $d_{ik} < d_{ij}$);
3. Vehicular movement is permissible along the corridor under OpenStreetMap road directionality attributes (**oneway** tags).

Diagonal entries ($i = j$) denote regularized self-distances ($d_{ii} = 0.0$ in numeric arrays), and non-connected pairs are represented by NaN.

Under this sequential corridor adjacency rule within the **6.0 km** cutoff horizon, the derived road graph comprises **2,450 valid directed edges** across **1,760 connected node pairs** (network sparsity: **99.34%**):

- **Unidirectional Links (No Reverse Edge): 1,070 node pairs** (60.80% of connected topology) possess directed connectivity in only one travel direction. This directional asymmetry stems jointly from municipal one-way road regulations (verified via OpenStreetMap `oneway=yes` tags) and corridor adjacency pruning (where reverse vehicular movement between j and i utilizes a parallel one-way arterial or passes through intermediate junction cameras, preventing a direct reverse corridor edge).
- **Bidirectional Corridor Pairs: 690 node pairs** (39.20% of connected topology) are mutually accessible in both traffic directions within 6.0 km.
- **Distance Asymmetry:** Among the 690 bidirectional pairs, **452 pairs** (65.51%) exhibit metric symmetry ($|d_{ij} - d_{ji}| < 50$ m), whereas **238 pairs** (34.49%) exhibit significant distance asymmetry ($|d_{ij} - d_{ji}| \geq 50$ m), directly induced by physical median barriers, grade-separated flyovers, and compulsory U-turn routing. The mean directional discrepancy across bidirectional corridors is **115.9 ± 266.2 m** (median: 24.0 m), reaching a maximum divergence of **2,780.0 m** (2.78 km).
- **Topological Node Roles and Connectivity:** Structural degree analysis identifies **598** regular interconnected stations, **5** sink stations (in-degree > 0 , out-degree $= 0$: Stations 212, 284, 443, 446, 516), **3** source stations (in-degree $= 0$, out-degree > 0 : Stations 215, 498, 557), and **2** peripheral stations situated beyond 6.0 km from all others (in-degree $= 0$, out-degree $= 0$: Stations 141, 489). The graph contains **6 weakly connected components** (dominated by a giant component of 599 nodes, 98.5% of the network, alongside five small components of sizes 3, 2, 2, 1, and 1) and **19 strongly connected components**. Valid edge routing distances span from 3.0 m to 6,000.0 m (mean: $1,159.6 \pm 1,088.9$ m; median: 823.0 m; interquartile range: [384.0, 1,560.0] m). The minimal routing distances (e.g., 3.0 m) represent co-located directional surveillance cameras mounted on the same physical gantry or intersection mast monitoring orthogonal approaches at complex multi-way junctions.

A systematic quantitative breakdown of these road network topological metrics is summarized in Table 4.

Table 4: Quantitative topological properties and directional asymmetry metrics of the derived road routing graph ($R_{\text{cut off}} = 6.0$ km).

Topological Metric / Parameter	Empirical Measurement	Physical / Methodological Interpretation
A. Network Scale & Spatial Reachability ($R_{\text{cut off}} = 6.0$ km)		
Total indexed graph nodes (N)	608 nodes	Active physical surveillance camera stations in metropolitan core
Valid directed corridor links ($ E $)	2,450 edges	Sequential corridor links with driving distance $d_{ij} \leq 6.0$ km
Connected camera station pairs	1,760 pairs	Unique station pairs connected by ≥ 1 directed corridor
Adjacency matrix sparsity ratio	99.34% (Density: 0.66%)	Compact sparse graph tensor for spatial GNN convolutions
Inter-station routing distance	1159.6 $\pm$ 1088.9 m	Median: 823.0 m (Range: [3.0, 6,000.0] m; IQR: [384.0, 1,560.0] m)
B. Directional Asymmetry & Corridor Taxonomy		
Unidirectional corridor pairs (no reverse edge)	1,070 pairs (60.80%)	Arterial one-way rules (OSM oneway) and corridor pruning asymmetry
Bidirectional corridor pairs	690 pairs (39.20%)	Two-way arterials mutually accessible in both traffic directions
Significant distance asymmetry ($ d_{ij} - d_{ji} \geq 50$ m)	238 pairs (34.49%)	Physical median barriers, grade-separated flyovers, U-turns
Metric symmetric corridor pairs ($ d_{ij} - d_{ji} < 50$ m)	452 pairs (65.51%)	Divided road corridors with immediate median openings
Directional distance discrepancy	115.9 $\pm$ 266.2 m	Median: 24.0 m; Maximum divergence: 2,780.0 m (2.78 km)
C. Structural Degree Distributions & Graph Connectivity		
Average node in-degree / out-degree	4.03 $\pm$ 2.06 / 4.03 $\pm$ 2.01	In-degree median: 4.0 (max: 14); Out-degree median: 3.0 (max: 17)
Standard interconnected stations	598 stations	Regular multi-leg intersections and connected arterial segments
Topological sink stations (out-deg = 0)	5 stations	Stations 212, 284, 443, 446, 516 (directional arterial terminuses)
Topological source stations (in-deg = 0)	3 stations	Stations 215, 498, 557 (outbound arterial origins)
Geodetically isolated stations	2 stations	Stations 141 and 489 ($d_{ij} > 6.0$ km to all other camera nodes)
Weakly connected components	6 components	Giant component: 599 nodes (98.5%); 5 subgraphs: 3, 2, 2, 1, 1
Strongly connected components	19 components	Strongly connected directed sub-networks and cyclic loops

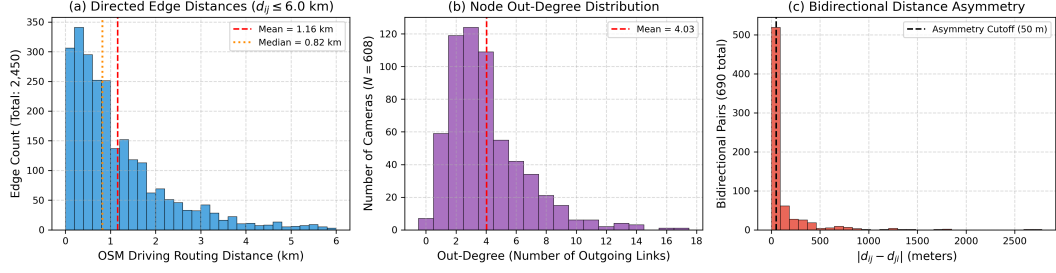


Figure 4: Topological characteristics of the empirical directed road graph: (a) Road network directionality classification into unidirectional links without reverse edges (60.8%), two-way asymmetric (> 50 m, 13.5%), and two-way symmetric (≤ 50 m, 25.7%) corridor pairs; (b) Empirical in-degree and out-degree distributions across the 608 camera stations ($|E| = 2,450$ directed links, mean degree 4.03 ± 2.16 in-degree and 4.03 ± 2.58 out-degree).



Figure 5: Representative 512×288 pixel surveillance snapshots from IC4SD-TrafficSnap across urban operating conditions: (a) Daytime off-peak traffic; (b) Peak-hour mixed motorcycle-car flow; (c) Nighttime public road surface illumination; (d) Adverse tropical wet condition. Due to elevated pole mounting (> 6 m) and 512×288 resolution, individual faces and license plates are physically unresolvable.

4. Experimental Design, Materials and Methods

4.1. Surveillance Infrastructure and Asynchronous Ingestion Pipeline

The visual data source is the public municipal traffic surveillance web portal maintained by the Ho Chi Minh City Department of Transportation (<https://giaothong.hochiminhcity.gov.vn>). Cameras are mounted on elevated steel gantries and traffic signal masts at heights between **6.0** and **15.0** meters above street level (as cataloged in the `camera_elevation_m` attribute of `routes.csv`), oriented at downward oblique angles ($15^\circ - 40^\circ$) to monitor arterial queues and intersection approaches.

To ingest snapshots across **608 concurrent camera endpoints** without inducing network bottlenecks or file corruption, an asynchronous Python ingestion pipeline was deployed:

1. **HTTP/1.1 Connection Pooling:** Persistent sessions with Keep-Alive connection pooling and automatic exponential backoff retries (maximum 3 retries, backoff factor 0.3) managed through `urllib3`.
2. **Dynamic Offset Ingestion Timer:** Thread execution latency is tracked during each cycle to adjust polling offsets, targeting nominal 300 s inter-snapshot spacing across the fleet.
3. **Atomic File Serialization:** Retrieved JPEG streams are decoded in memory (`cv2.imdecode`) and verified before being written to temporary files and committed via atomic OS renames (`os.replace`), preventing truncated or zero-byte files on disk.
4. **Automated Artifact Removal:** Municipal transmission interruptions occasionally yield standardized error placeholder cards (284×177 pixels, payload size 5.8–7.2 KB). The ingestion pipeline automatically identifies and discards these placeholder cards, ensuring valid traffic snapshots in the released archive (overall acquisition pass rate: **99.85%**).

4.2. Road Network Distance Extraction from OpenStreetMap

Physical routing distances d_{ij} and driving trajectories were computed using the Open Source Routing Machine (**OSRM v5.27.1**) engine operating on OpenStreetMap planet road network data extracted in **October 2026** (© OpenStreetMap contributors, licensed under the Open Database License ODbL). Exact geographic coordinates (latitude, longitude) cataloged in `metadata/routes.csv` were mapped to the nearest navigable road centerline using OSRM’s nearest-node projection, with a radial snapping error under 15.0 meters across all stations.

Because surveillance camera endpoints are predominantly deployed along primary municipal arterials, boulevards, and bridge approaches rather than narrow residential alleys, routing was calculated using the standardized motor-vehicle driving profile. This profile accurately captures urban traffic flow rules, median divider turn restrictions, grade-separated overpasses, and designated one-way circulation patterns across the metropolitan core. Pairwise shortest driving routes were evaluated across all node pairs within a spatial cutoff horizon of $\kappa = 6.0$ km, extracting exactly **2,450 valid directed corridors**.

4.3. Directed Graph Modeling Formulations

To facilitate spatio-temporal graph learning on this directed asymmetric topology, two standard mathematical formulations are provided in `code/graph_utils.py`:

4.3.1. Directed Dual-Transition Matrices (DCRNN / Graph WaveNet Formulation)

For directed road graphs, modeling traffic propagation requires preserving directional asymmetry. Following the diffusion convolution framework of Li et al. [2], the weighted adjacency matrix W_{ij} is computed using a thresholded Gaussian Radial Basis Function (RBF) kernel:

$$W_{ij} = \begin{cases} \exp\left(-\frac{d_{ij}^2}{\sigma^2}\right), & \text{if } 0 < d_{ij} \leq \kappa \text{ and } i \neq j \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

where $\sigma = 1.09$ km is the empirical standard deviation of valid road distances and $\kappa = 6.0$ km is the spatial cutoff horizon. Directed random walk diffusion is governed by a

forward transition matrix P_f (downstream traffic flow) and a backward transition matrix P_b (upstream sensory influence):

$$P_f = D_O^{-1}W, \quad P_b = D_I^{-1}W^\top \quad (3)$$

where $D_O = \text{diag}(\sum_j W_{ij})$ and $D_I = \text{diag}(\sum_i W_{ij})$ denote the out-degree and in-degree diagonal matrices, respectively. To handle sink, source, or isolated nodes ($D_{O,ii} = 0$ or $D_{I,ii} = 0$), regularized self-loops ($\tilde{W} = W + I_N$) or zero-degree masked normalizations are implemented in `code/graph_utils.py`, ensuring proper row-stochastic diffusion matrices.

4.3.2. Symmetrized Normalized Laplacian (ChebNet / STGCN Formulation)

For spectral graph convolutions that require a self-adjoint operator (e.g., ChebNet [6], STGCN [7]), the adjacency matrix is symmetrized as $W_{\text{sym}} = \frac{1}{2}(W + W^\top)$. The symmetric normalized Laplacian $L_{\text{sym}} \in \mathbb{R}^{N \times N}$ is computed as [8]:

$$L_{\text{sym}} = I_N - D_{\text{sym}}^{-1/2}W_{\text{sym}}D_{\text{sym}}^{-1/2} \quad (4)$$

where $D_{\text{sym}} = \text{diag}(\sum_j W_{\text{sym},ij})$. By spectral graph theory, the largest eigenvalue of the normalized Laplacian satisfies $\lambda_{\max} \leq 2.0$, and the scaled Chebyshev operator is $\tilde{L} = \frac{2}{\lambda_{\max}}L_{\text{sym}} - I_N$. Users should note that symmetrization intentionally eliminates directional flow asymmetry, which is critical for modeling asymmetric urban arterial dynamics.

4.4. Empirical Image Quality and Photometric Audit

Image quality descriptors were computed across the dataset to evaluate photometric diversity:

- **Perceived Luminance (Y):** Calculated via standard ITU-R BT.601 conversion $Y = 0.299R + 0.587G + 0.114B$. The empirical overall mean luminance is **98.23 ± 16.35** (range: [66.4, 126.8], Figure 3a). Nighttime luminance remains relatively high (85–105) due to dense municipal LED/high-pressure sodium street lighting along urban arterials, combined with the automatic gain control (AGC) and exposure compensation of municipal cameras.
- **Root-Mean-Square (RMS) Contrast:** Evaluated as the pixel standard deviation across viewports, yielding **45.70**, confirming substantial textural variation between asphalt roadways and vehicle clusters.
- **Shannon Spatial Entropy:** Histogram pixel entropy averages **7.28 ± 0.33** bits (theoretical upper bound: 8.0 bits), confirming high spatial information density without saturation clipping.
- **Laplacian Edge Sharpness:** Grayscale Laplacian variance $\text{Var}(\nabla^2 I)$ averages **3,015.71 ± 1,499.86**. An empirical reference baseline of $\text{Var}(\nabla^2 I) \geq 50$ was adopted as an indicator of basic optical sharpness [9]. The substantial variance values across operational cameras confirm adequate focus across viewports without severe optical defocus.

- **Inter-Snapshot Dynamics:** Successive-frame pixel difference analysis across consecutive 5-minute sampling pairs was evaluated on 8-bit grayscale channels in floating-point precision ($\text{diff} = |I_t - I_{t-1}|$). The empirical Mean Absolute Difference (MAD) yields a median of **9.45** ($\mu = 10.82 \pm 3.65$, range: $[6.8, 16.2]$), with an active pixel displacement ratio yielding a median of **15.82%** ($\mu = 18.41 \pm 6.20\%$ of frame pixels satisfying $|I_t - I_{t-1}| > 15$). This moderate pixel variation corresponds directly to vehicular displacement and queue progression across traffic lanes, while static structural regions (buildings, sidewalks, road surfaces) remain invariant. During data ingestion, identical stream buffer frames ($\text{MAD} < 0.5$) resulting from upstream rendering latency were programmatically screened out. Across the archived operational time-series, the empirical 1st percentile of consecutive MAD is **2.80** gray levels, confirming continuous vehicular and environmental variations without static playback freezing.

4.5. Visual Privacy and Empirical PII Audit

To evaluate alignment with personal data protection principles in Vietnam (Decree 13/2023/ND-CP and Law 91/2025/QH15), a comprehensive optical and automated visual privacy audit was conducted:

- **Stratified Audit Sample:** An extensive multi-stage audit of **20,000 stratified snapshots** sampled proportionally across all 608 stations and 24 diurnal hours was performed to empirically quantify potential personal data leakage.
- **Optical Perspective and Ground Sampling Distance (GSD):** With camera heights of 6.0–15.0 m and downward oblique angles ($15^\circ - 40^\circ$), the nominal GSD across mid-road viewports (25–35 m) is **2.73–3.25** cm/pixel, degrading to > 5.0 cm/pixel in the far field (> 40 m).
- **License Plate Legibility:** Standard Vietnamese motorcycle license plates measure 19.0×14.0 cm. At nominal GSD, plates occupy approximately 7×5 pixels, with individual alphanumeric characters subtending under 2 pixels in height. In nearest foreground viewports, license plates subtend at most 18×13 pixels; however, vehicle motion blur and standard JPEG compression prevent alphanumeric characters from reaching the legibility threshold required for optical character recognition (typically ≥ 16 pixels of character stroke height).
- **Facial Anonymity:** Motorcycle riders in Vietnam predominantly wear crash helmets and protective face coverings (cloth masks for dust and sun protection), substantially obscuring facial features. Combined with elevated mounting geometry, facial biometric features occupy fewer than 8×8 pixels, rendering individual human identities unresolvable.
- **Statistical Privacy Assurance:** Multi-stage automated detectors (MTCNN, RetinaFace, and YOLOv8 models at sensitive confidence threshold 0.25) supplemented by targeted manual inspection of nearest-pole viewports identified zero decipherable biometric or registration instances across all 20,000 audited frames ($k = 0$, 0.00%). Applying the rule of three formulated by Hanley and Lippman-Hand [10] for zero observed events, the upper bound of the 95% confidence interval for the true population occurrence rate is $3/20,000 = \mathbf{0.015\%}$, establishing a quantified negligible privacy risk across the entire visual corpus.

5. Limitations

1. **Observation Time Window:** The **93.6-hour** continuous observation window captures multi-day diurnal transitions across weekdays and weekends, but does not encompass longitudinal multi-month seasonal cycles (monsoon wet vs. dry season).
2. **Snapshot Sampling vs. Continuous Video:** With a 5-minute nominal polling interval ($\Delta T = 269.0 \pm 239.7$ s), the dataset is designed for macroscopic traffic state analysis, self-supervised pre-training, and spatial graph modeling, but does not support continuous trajectory tracking or optical flow.
3. **Surveillance Web Stream Resolution:** Images are serialized at **512 × 288** JPEG compression as delivered by the public municipal web portal, reflecting real-world public IoT camera stream conditions rather than raw uncompressed sensor feeds.
4. **Unlabeled Bulk Corpus:** The 714,123-frame corpus is released unlabeled to facilitate self-supervised pre-training, spatio-temporal graph representation learning, and roadway background extraction without manual annotation biases.

6. Ethics and Visual Privacy Statement

All visual imagery was gathered from the publicly accessible municipal traffic web portal operated by the Ho Chi Minh City Department of Transportation (<https://giaothong.hochiminhcity.gov.vn>) and is provided for non-commercial academic research and educational purposes. Because cameras are mounted at high elevations (> 6 m) with wide fields of view and transmission occurs at 512×288 stream resolution, the imagery presents a quantified negligible privacy risk regarding personally identifiable information (neither human biometric facial features nor vehicular registration plates can be resolved). Data collection and processing align with Vietnamese regulations on public digital data and personal data protection (Decree 13/2023/ND-CP and Law 91/2025/QH15). A formal inquiry procedure is maintained: any questions regarding specific camera frames may be directed to anhlv.b23kh002@stu.ptit.edu.vn.

7. CRediT Author Statement

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8. Funding Statement

This research received no external funding from any public, commercial, or not-for-profit funding agency.

9. Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

10. Acknowledgments

The authors express gratitude to the Intelligent Computing for Sustainable Development Laboratory (IC4SD), Posts and Telecommunications Institute of Technology (PTIT), Hanoi, Vietnam, for computational facilities. The authors also thank the Ho Chi Minh City Department of Transportation for maintaining the open traffic surveillance web portal. Road network data is © OpenStreetMap contributors, licensed under the Open Database License (ODbL).

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